Nayoung Ku

+82 10-8131-0702 nayoungku1@gmail.com nayoung-ku nayoungku1 nayoungku1.github.io

Education

Handong Global University Feb. 2023 – Feb. 2026
B.Sc. in Life Science and Applied Artificial Intelligence (cum laude, GPA: 4.0/4.5) *Pohang, South Korea*
- **Bachelor's Thesis:** Language-Model-Based RNA-seq Data ETL Workflow for Ovarian Cancer Classification Model

National Institute for Lifelong Education Feb. 2021 – Feb. 2023
B.L.A. in Psychology, Bachelor's Degree Examination for Self-Education (GPA: 4.0/4.3) *Seoul, South Korea*

Honors & Awards

Outstanding Performance Scholarship (Top 30%) Aug. 2025
Big Data Global Research Program – Big Data COSS Consortium, Ministry of Education

Big Data Global Research Program Fellowship (1 of 3 selected from HGU) Apr. 2025
Funded summer research at UNLV – Big Data COSS Consortium, Ministry of Education

3rd Place, Daewoong Pharmaceutical President's Award (out of 900 teams) Oct. 2024
JUMP AI Drug Discovery Challenge – Organized by KPBMA & Sponsored by the Ministry of Health and Welfare

Big Data Seagull Scholarship Spring 2024, Fall 2024
Achievement-based scholarship for big data/AI coursework – certified by HGU School of Applied Artificial Intelligence

Research Experience

Undergraduate Researcher → Researcher Aug. 2024 – Present
Handong Global University – Advisor: Prof. Jaeyoung Chun *Pohang, South Korea*
- Exploring gene regulatory network-informed approaches to predict **transcriptomic shifts** in response to **genetic perturbation (CRISPRi)**, aiming for zero-shot prediction on unseen cell lines
- Standardizing heterogeneous **Perturb-seq** datasets and building reproducible model training/evaluation pipelines
- Developed a diffusion-based probabilistic demultiplexing pipeline for hashed scRNA-seq data that learns the underlying signal distribution without hard-threshold cutoffs
- Constructed a high-quality annotated training dataset for **circulating tumor cell (CTC)** detection in unstained blood **microscopy images** and conducted a systematic literature review on CTC morphology
- Initiated and led an **IC50 prediction project** for IRAK4 inhibitors from chemical structure (SMILES), generated molecular descriptors, and designed feature selection methods for ML regression models
- Achieved **top 3 finalist** status at a national AI drug discovery challenge (see Honors & Awards) and **presented the poster at AI Pharma Korea Conference 2024**

Summer Undergraduate Research Trainee Jun. 2025 – Jul. 2025
University of Nevada, Las Vegas – Advisor: Prof. Mingon Kang (DataX Lab) *Nevada, United States*
- Developed a lightweight **patch tiling and filtering algorithm** for **whole slide images (WSIs)** from the **TCIA CMB-LCA**, reducing raw data volume by **80%** (128GB → 25GB), significantly lowering computational overhead
- Built a hybrid deep learning framework integrating **Vision Transformer** embeddings with an **LSTM**-based classifier for large-scale, imbalanced tumor classification across WSIs

Undergraduate Researcher Jan. 2025 – Apr. 2025
Seoul National University – Advisor: Prof. Howard Lee (CCADD) *Seoul, South Korea*
- Contributed to an **NLP research project commissioned by Korea's Ministry of Food and Drug Safety (MFDS)**, building a **Korean–English parallel sentence corpus** of pharmaceutical regulatory documents
- Standardized data collection, annotation, and management protocols across a multi-source document corpus, establishing a reusable curation pipeline for the research team, and designed a sentence alignment algorithm for automated parallel corpus construction from heterogeneous document sources

Undergraduate Research Intern Mar. 2024 – Nov. 2024
Handong Global University – Advisor: Prof. Taejin Ahn (BioData Lab) *Pohang, South Korea*
- Analyzed clinical **Tumor-Educated Platelet (TEP) RNA-seq** datasets to identify **ovarian and colorectal cancer biomarkers**, applying feature selection and hierarchical clustering to isolate signal from high data sparsity
- Developed a **transformer-based preprocessing framework** encoding raw RNA sequences of TEPs directly into embeddings without traditional count-matrix construction and achieved **AUC 0.71** in ovarian cancer classification

Presentations

N. Ku, J. Kim. "No Cell Left Behind: Recovering Discarded Cells via Self-Supervised Diffusion Denoising." *Bio-AI Capstone Design Fair*, Handong Global University, Pohang, South Korea, Dec. 2025.

N. Ku. "Diagnosing Cancer with Genomic Context." *Capstone Design in Life Science Fair*, Handong Global University, Pohang, South Korea, Nov. 2024.

N. Ku, S. Kwon, Y. Choi. "IRAK4 IC50 Inhibition Prediction." *AI Pharma Korea Conference 2024*, Seoul, South Korea, Oct. 2024. (Poster; invited as a winning team of the JUMP AI Drug Discovery Challenge)

Teaching Experience

Introduction to AI and Machine Learning, *Teaching Assistant* Aug. 2025
Addis Ababa Science and Technology University – Instructor: Prof. Dai Yon Cho *Addis Ababa, Ethiopia*

Big Data Modeling and Platforms, *Teaching Assistant* Mar. 2025 – Jun. 2025
Handong Global University – Instructor: Prof. Jaeyoung Chun *Pohang, South Korea*

AI Application Programming, *Teaching Assistant* Mar. 2025 – Jun. 2025
Handong Global University – Instructor: Prof. Sangsan Lee *Pohang, South Korea*

Technical Skills

Languages: Python, R, SQL, Bash
ML/DL: PyTorch, scikit-learn, Weights & Biases, MLflow, Optuna, HuggingFace
Bioinformatics: Scanpy, scRNA-seq Analysis, Bulk RNA-seq, Histopathology & Microscopy Image Analysis
Infrastructure: Git, GitHub, Slurm, LSF, MIG-partitioned GPU clusters, Linux

Community Involvement

R Programming Camp, *Teaching Assistant* Jan. 2026
Mentored high school students in data analysis and visualization using R *Pohang, South Korea*

Handong Office of International Affairs, *Exchange Student Mentor* Aug. – Dec. 2024
Assisted exchange students with adaptation to Korean culture and residential college *Pohang, South Korea*

Incheon Beauty and Art High School Python Camp, *Teaching Assistant* Aug. 2023
Supported Python programming instruction for elementary to high school students *Incheon, South Korea*